Financial Dashboard



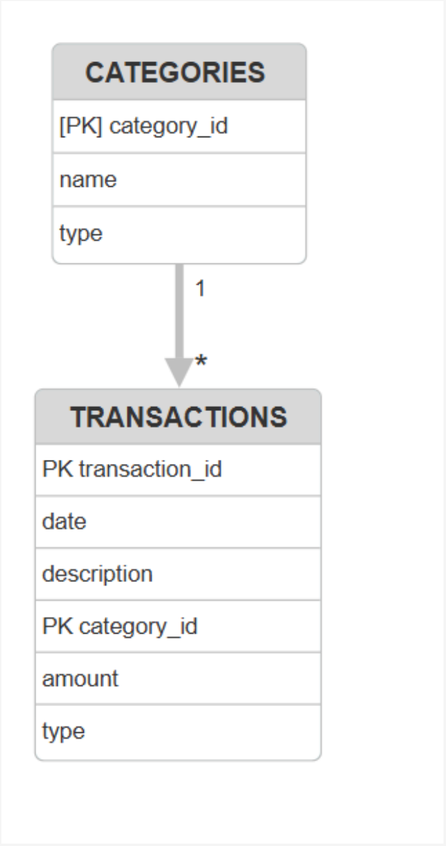
Finance Analytics Dashboard – GitHub Repository

Development/ Reflection phase
DLBITPEWP01_E

Data model and system architecture

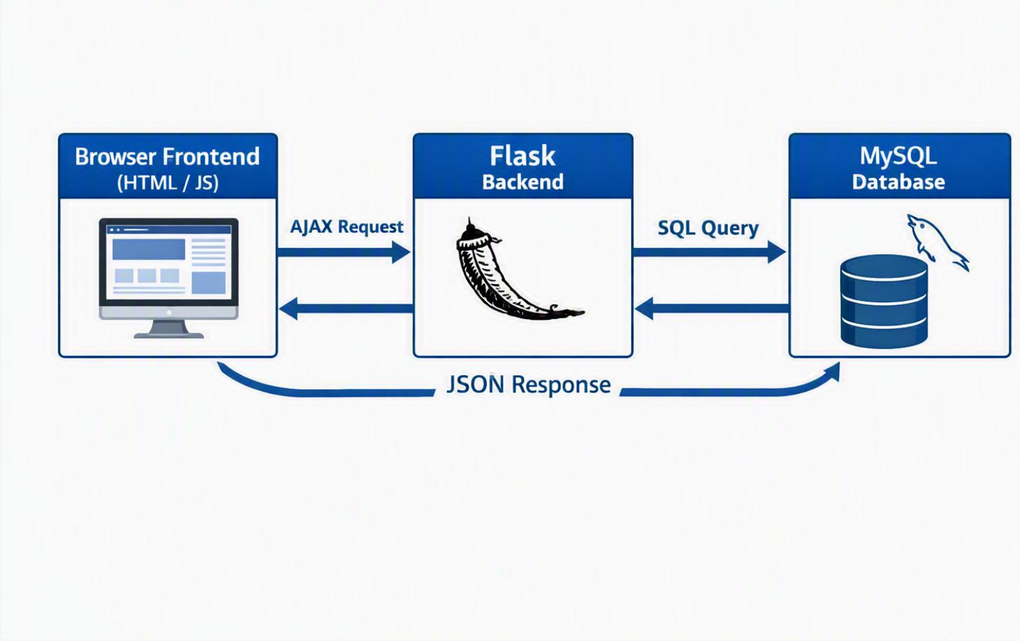
Database:

- Relational MySQL database
- Categories and transaction tables
- One to many relationships between categories and transactions
- Transactions linked to categories using category_id
- Income and expense types supported



System architecture:

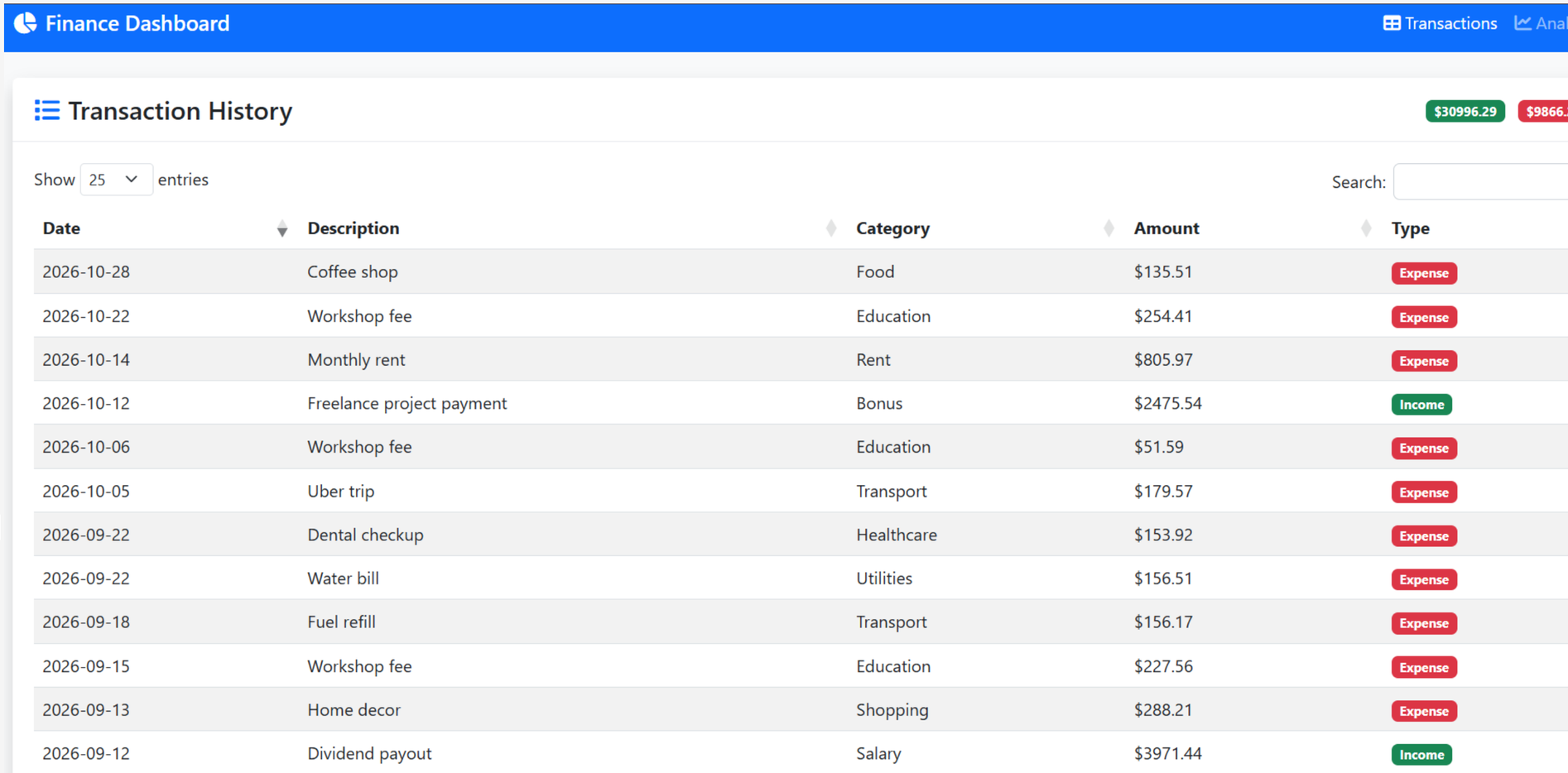
- The application follows a client-server architecture, the frontend uses the returned data to populate transactions and financial visualizations



Analytics Table

The finance analytics dashboard includes an interactive transaction table, displaying financial records in a structured format. Implemented features shown include:

- Transaction date
- Description
- Category
- Amount
- Transaction type
- Search functionality
- Column sorting
- Adjustable number of displayed entries
- Income expense indicators



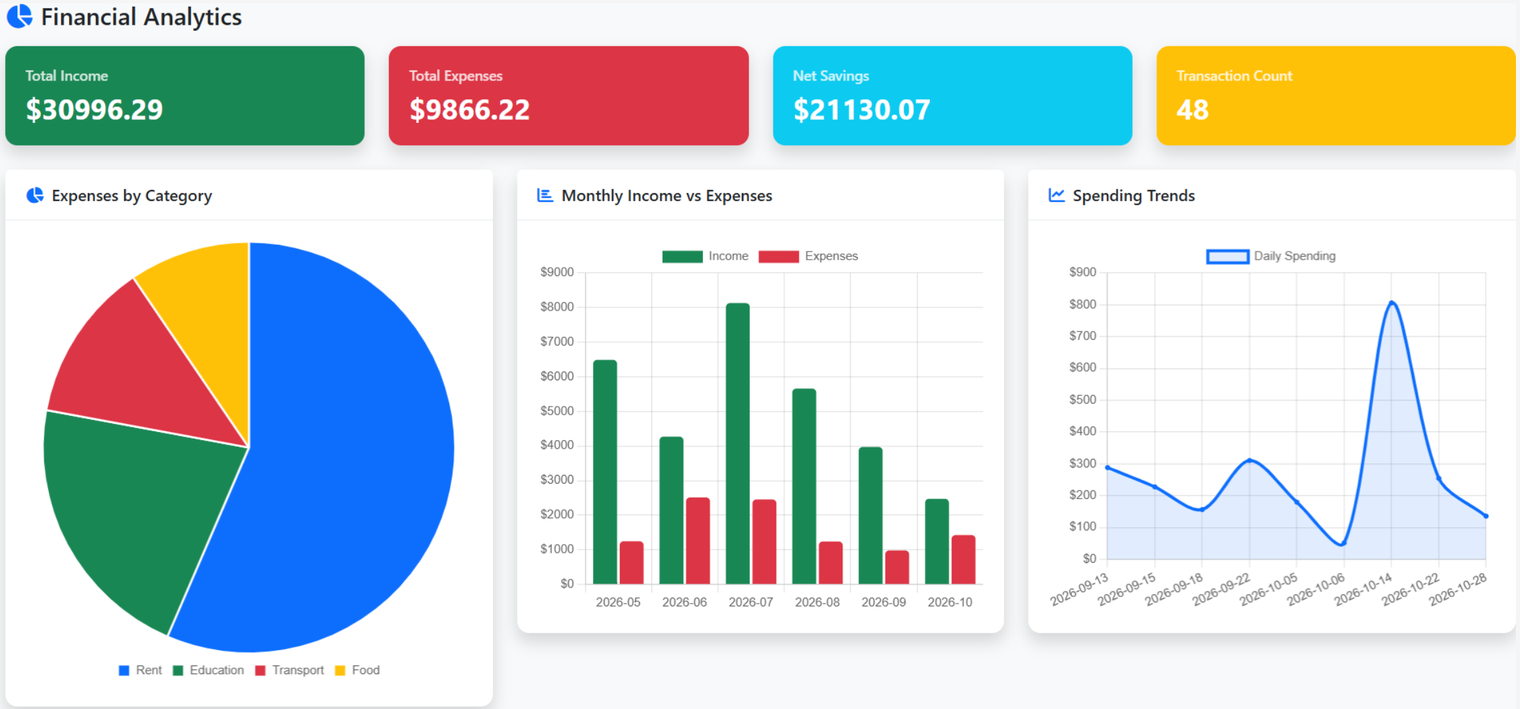
Date	Description	Category	Amount	Type
2026-10-28	Coffee shop	Food	$135.51	Expense
2026-10-22	Workshop fee	Education	$254.41	Expense
2026-10-14	Monthly rent	Rent	$805.97	Expense
2026-10-12	Freelance project payment	Bonus	$2475.54	Income
2026-10-06	Workshop fee	Education	$51.59	Expense
2026-10-05	Uber trip	Transport	$179.57	Expense
2026-09-22	Dental checkup	Healthcare	$153.92	Expense
2026-09-22	Water bill	Utilities	$156.51	Expense
2026-09-18	Fuel refill	Transport	$156.17	Expense
2026-09-15	Workshop fee	Education	$227.56	Expense
2026-09-13	Home decor	Shopping	$288.21	Expense
2026-09-12	Dividend payout	Salary	$3971.44	Income

Financial visualizations

The dashboard provides three visualizations for analyzing financial activity:

1. Expenses by category: A pie chart showing the distribution of expenses across categories
2. Monthly income vs expenses: Bar chart comparing income and expenses across months
3. Spending trends: Line chart displaying spending patterns over time

The dashboard also displays summary indicators for total income, total expenses, net savings and transaction count



Technology stack and development changes

Component	Technology
Frontend	HTML5, CSS3, JavaScript
Framework	Bootstrap 5
Tables	Data Tables.js
Charts	Chart.js
Backend	Python 3 + Flask
Database	MySQL
API	RESTful API + JSON
Testing	Postman
Version control	Git + GitHub

The technology stack remained consistent, the structure and visual design of the dashboard evolved significantly during development.

- The original concept emphasized two primary views, a transaction listing and a visualization page. During development these ideas were developed into a more integrated financial analytics dashboard. The layout and overall design were changed to provide a more cohesive dashboard experience.
- Summary cards for total income, total expenses, net savings and transaction count
- The positioning and presentation of the financial information were refined to make important information immediately available

Development challenges



Challenges encountered include:

1. Database connection – Establishing a reliable communication between the Flask application and MySQL database to ensure he application would retrieve the required financial data correctly
2. API integration – Connecting the frontend to the Flask REST API and ensuring that API requests and responses worked correctly with dashboard components
3. Code errors – Errors occurred during development while different components were being implemented and connected, these required troubleshooting and repeated testing
4. Component integration – The project required the frontend, backend, database, DataTables.js and Chart.js to work together as one application.

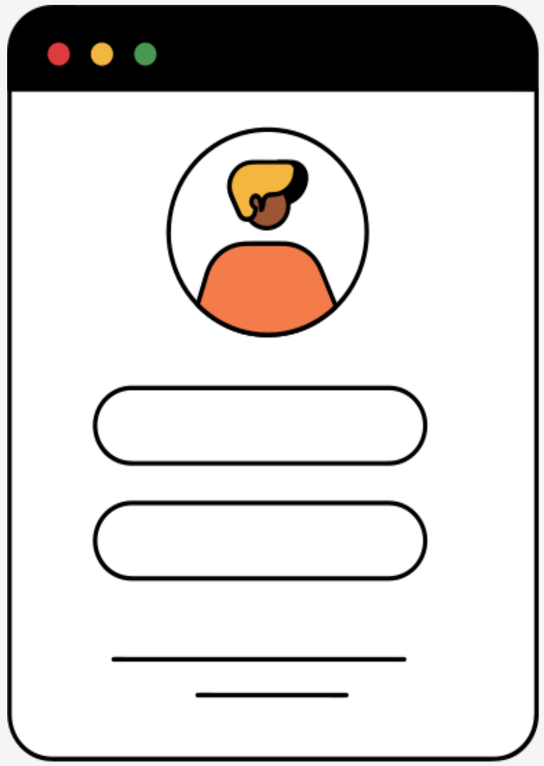
How were these challenges addressed? :

- Components were developed and tested iteratively
- Errors were investigated and corrected during implementation
- Database and API communication was refined
- The application was repeatedly tested until the different components worked together successfully.

Improvements made and features added

Features added:

- Financial summary cards
- Refined dashboard structure and visual design
- Interactive transaction table
- Search functionality
- Sorting functionality
- Adjustable table entries
- Spending trends visualization



Improvements made:

- Implemented the planned transactions
- Implemented all three planned financial visualizations
- Added financial summary cards for an immediate overview of financial activity
- Refined the dashboard layout and visual design
- Integrated frontend, backend and database components into functional application
- Improved analytical visibility
- Improved data visualization

Development and learning

The development process has demonstrated the importance of integrating individual technologies into a complete working system. Working with the database, API, frontend and visualization components required iterative testing, troubleshooting and refinement. Resolving database connectivity issues, API integration, problems and code errors provided practical experience in developing and integrating a full stack web application



Current status and steps

The finance analytics dashboard has reached a functional development stage. The completed prototypes include:

- Interactive transaction table
- Financial summary cards
- Expenses-by-category pie chart
- Monthly income vs expenses bar chart
- Spending trends line chart
- Flask backend
- MySql backend
- REST API integration
- Integrated dashboard layout



Next steps:

The next stage will have a focus on strengthening the dashboard for real world use by implementing authentication, expanding filtering and reporting capabilities, improving data persistence and deploying the application to a cloud environment. These enhancements would transform the current functional dashboard into a more complete financial analytics solution